Shane Reynolds Portfolio

2027

Wall Light - Design Focused



Path Light - Cost Optimized



Wall Lights - Thoughtfully Engineered



Bachelor of Industrial Product Design Engineering



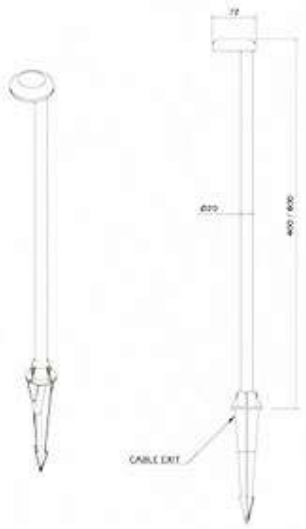


ARX-724 “POLO”

POLO is a CNC machined brass/aluminium, IP65 garden path light designed for high-end residential landscape/architectural applications.

Development Period	~4 Months 2025-2026	
Company / Brand	ARX / Aqualux	ARX /
Product Status	Fully Released	500+ Sold

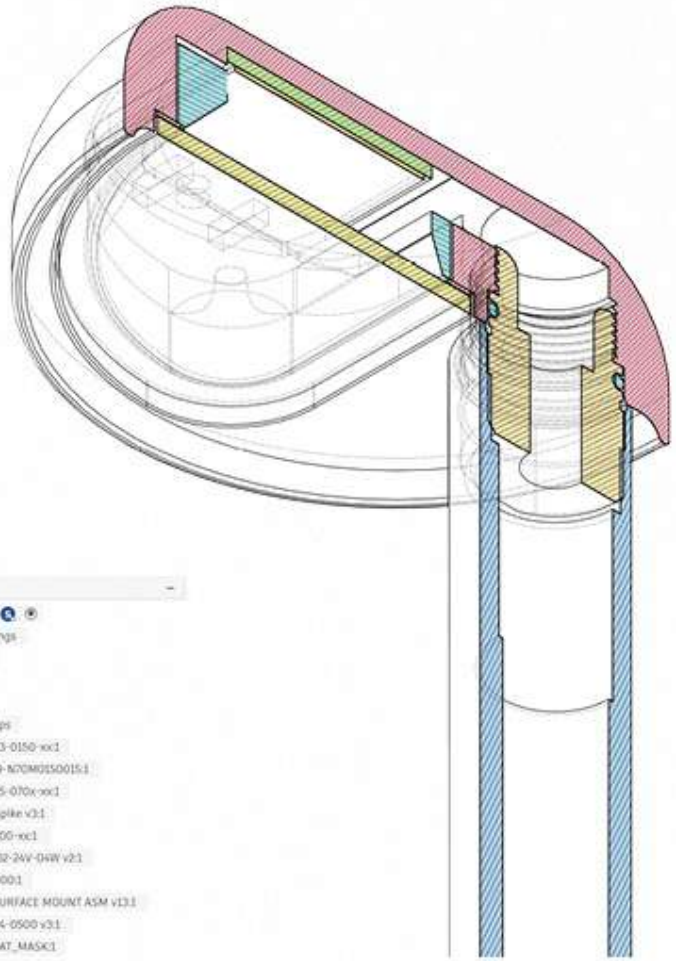
Brief Definition
Concept Development
CAD Modelling (Fusion)
Ingress Protection Design
Design for Manufacture
Design for Assembly
Manufacturing Documentation
AU/CN CNC Coordination
BOM management & Costing
Odoo Product Documentation
Webpage Management
Product Rendering
Product Photography

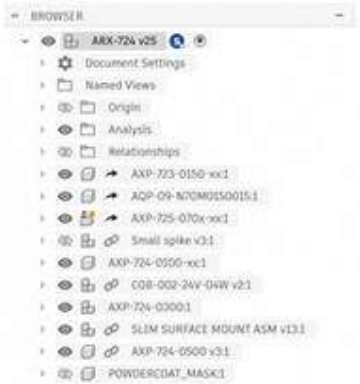


POLO Brief

Design a cost optimised, IP65 garden path light available in brass and powder coated aluminium for the ARX and Aqualux Lighting brands.

Disrupt the market for brass landscape lighting with reliability, price point, and aesthetic excellence.









Constraints



POLO was constrained heavily by mechanical, electrical, and costing requirements. Many iterations were required to refine a manufacturable solution that could house the electronics and IP sealing components in a sleek, visually pleasing package.

When constraints are in place for a design project, CAD becomes the primary development domain. True designers and true engineers design for function in CAD where specifications are defined.

Yes, industrial designers are taught to sketch first; however, that methodology is like eating desert before the main course, in my opinion. Get it right, then make it nice.

POLO was constrained heavily by mechanical, electrical, and costing requirements.

- Minimum Material Usage
- Minimum Machining Cycle Time
- 4W LED Thermal Dissipation
- Maximum Electronics Cavity Accessibility
- Minimum Seals Count
- Maximum IP65 Seals Reliability
- Maximum Aesthetic Appeal





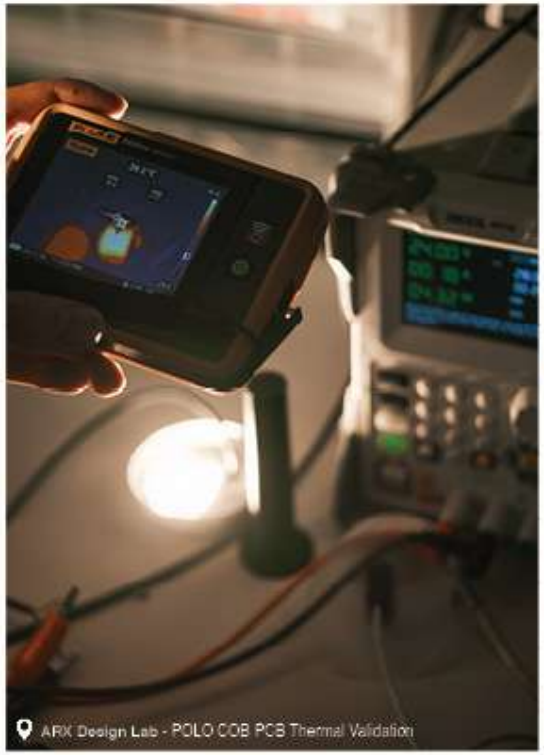
Testing

Polo was planned and costed in a bill of materials spreadsheet prior to production investment.

The ERP integration of POLO took place in Odoo management software. Pard codes, a bill of materials, routing flow, and attribute/variant configuration was setup to ensure the ordering pipeline was stable.

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ARX Design Lab - POLO COB PCB Thermal Validation



ARX Design Lab - POLO IES Testing and Optical Validation



ARX Design Lab - POLO Technical Drawings and Machined Samples Review

DFM & DFA Optimisation

POLO was designed for both in-house and external CNC manufacturing. Fully detailed CAD files and technical drawings were produced with appropriate GD&T to achieve a good metalwork cost to finish ratio.

POLO was designed for both in-house and external awhstj ry sr6jkd ugkra6u 4wy xf r5u dr6u













Documentation

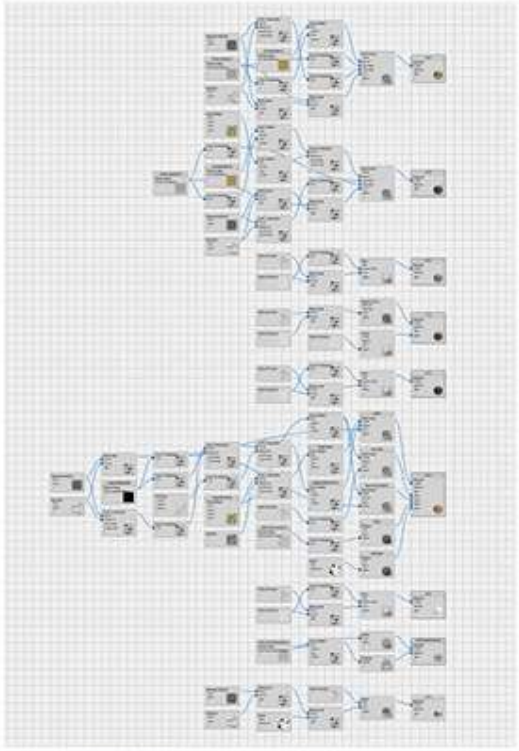
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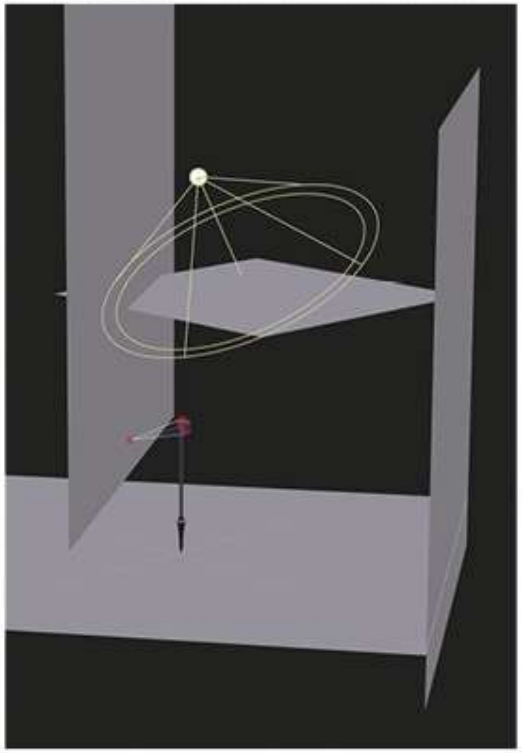
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ARX Design Lab - Fibre Laser Engraved POLO Component Sample for Archival







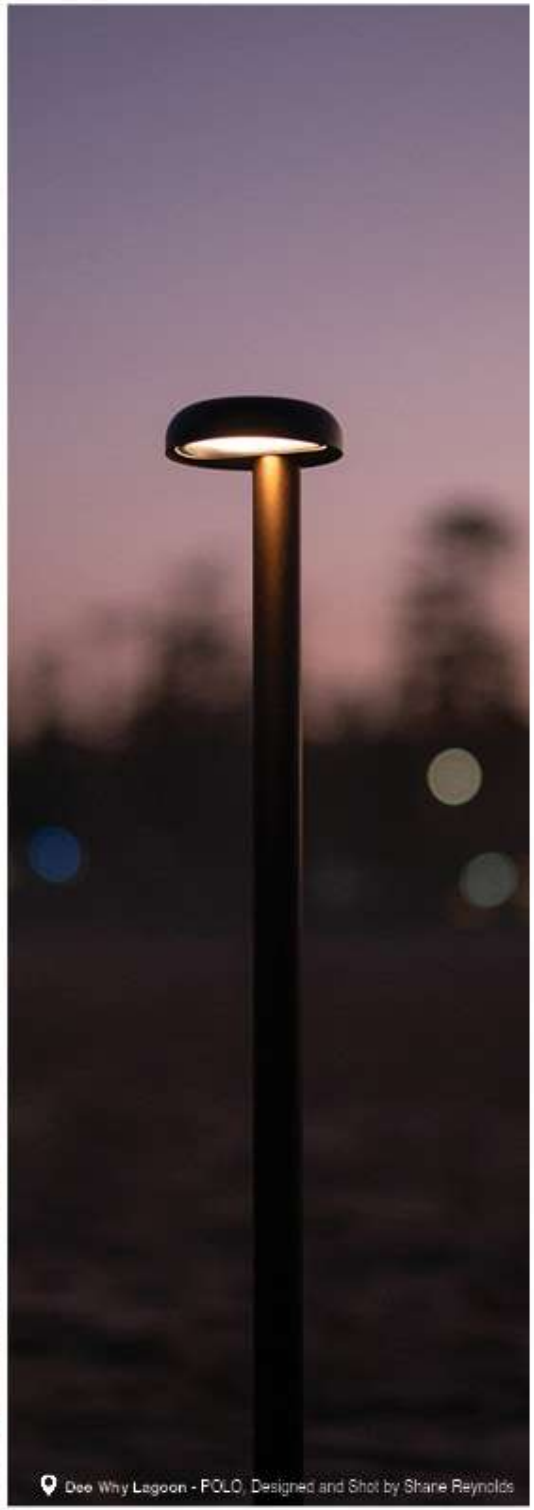


KeyShot









Dee Why Lagoon - POLO, Designed and Shot by Shane Reynolds



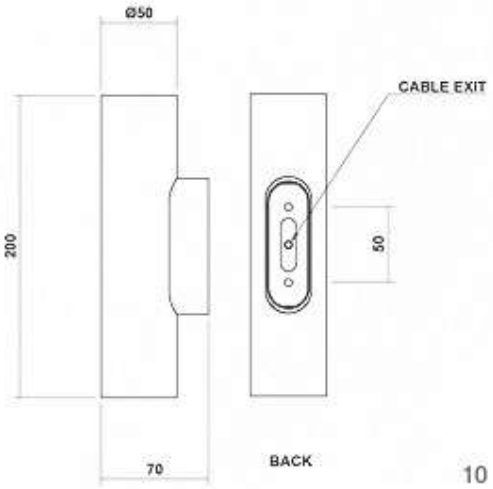


ARX TULA FAMILY

The TULA family is a range of CNC machined brass/aluminium, IP65 architectural lights designed for high-end residential architectural applications.

Development Period	~6 Months 2024-2025	
Company / Brand	ARX	ARX
Product Status	Fully Released	300+ Sold

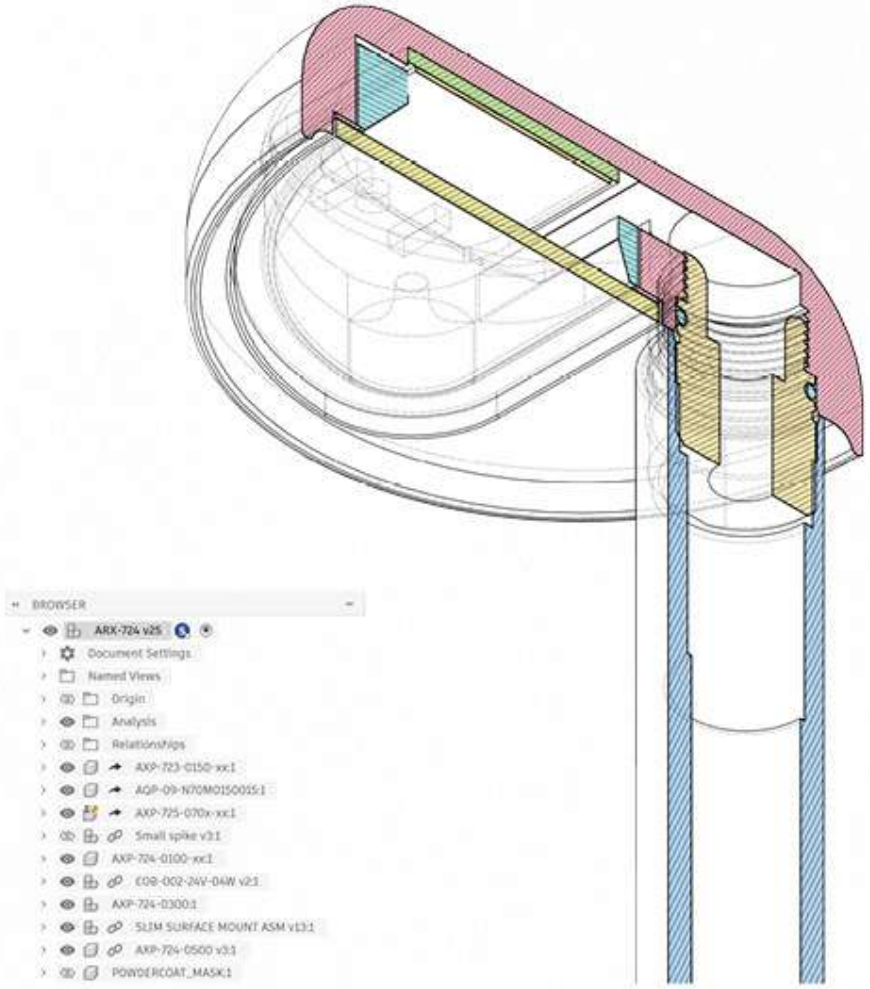
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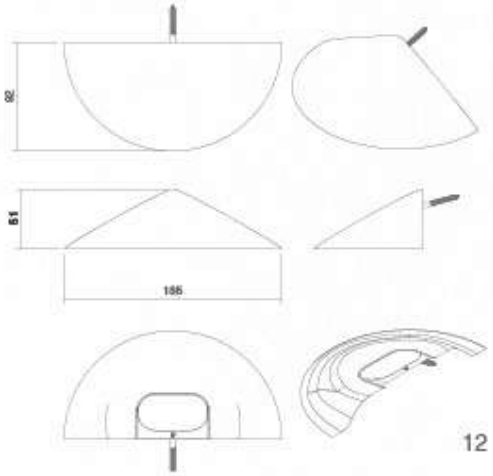


Kina Wall Sconce

Kina is a cast, concrete wall sconce that provides a smooth light wash for subtle and artistic lighting applications in a refined package.

Development Period	~2 Months 2026
Company / Brand	Pho (Personal Brand)
Product Status	Functional Concept

Design Philosophy
Concept Development
Form Optimisation
Mould Design
FFF Prototyping
Silicone Casting
Plaster & Concrete Casting
Functional Prototype
Product Photography

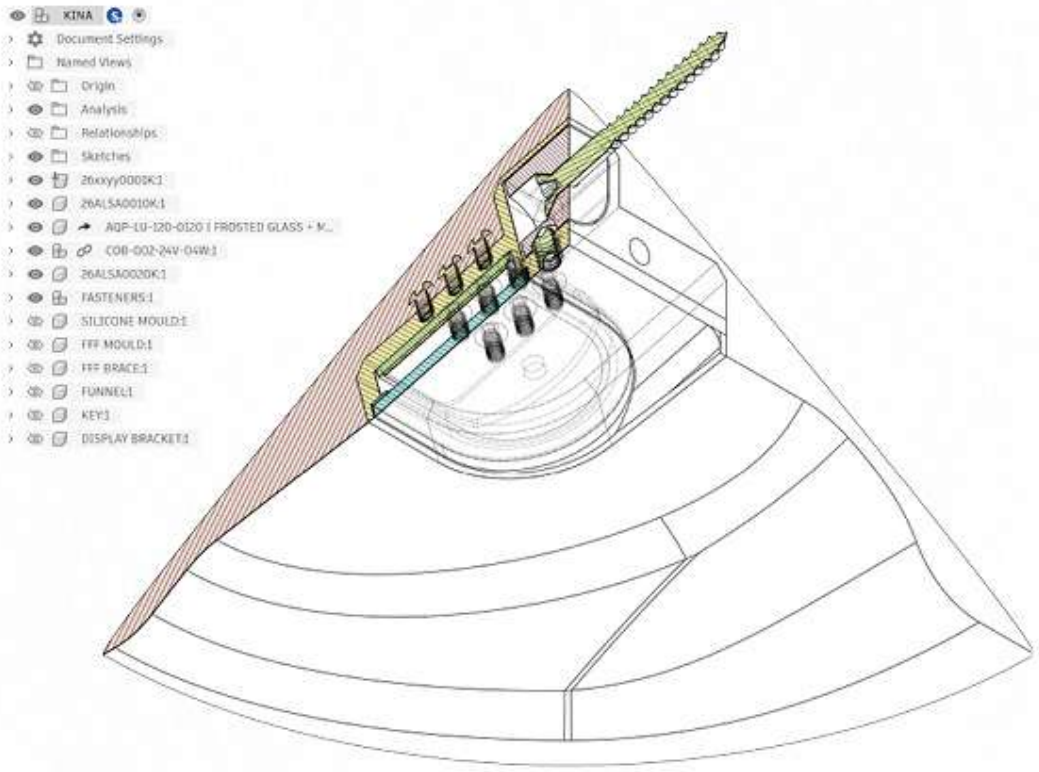


Kina Philosophy

In New Zealand where I grew up, we call sea urchin "kina." Kina have always interested me as they are rather simple creatures. During my lifetime, the kina population exploded in New Zealand because of the right environmental conditions.

There's a beauty in how the success of kina was not their choice, at least from the human perspective. Contemporary, individualist, western thought tells us that maximal autonomy and an internal locus of control are what we should strive for; however, I think there is just as much power in the surrender to what you can not control. Just as kina did, and still do.

I like to think that I designed Kina because of the right environmental conditions.





Prototyping

Kina was initially prototyped using plaster of paris and FFF printed 2-part moulds. This method sufficed during the refinement of the conoid form, but a higher complexity mould would later be designed to enable more complex geometric features and increase yield rate of finished Kina components.

It was important to me to keep the prototyping process for Kina fun. I primarily worked on Kina after work and was more playful in my approach as a result. Methodological approaches have their place, but intuitive, experience based design is also valid in some circumstances.





Mould Design and Simulation*

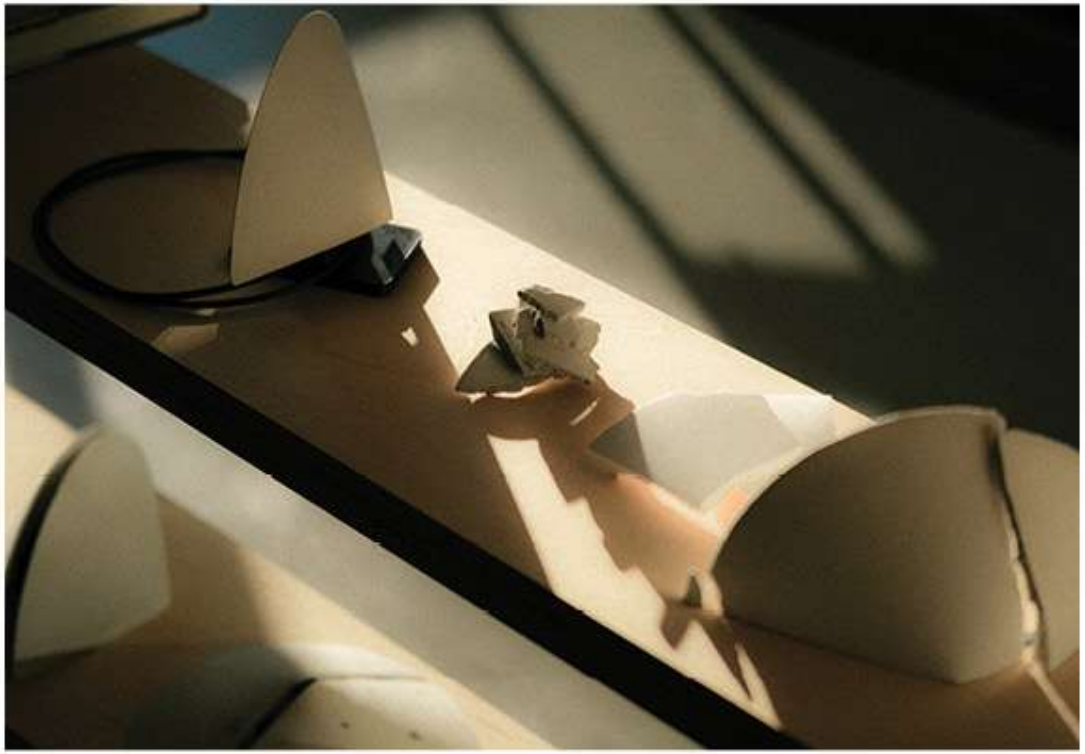
When the form of Kina was roughly decided, I moved to working with concrete. I developed a 1-part butterfly silicone mould for the primary positive cavity and a 2-part bracing around the silicone core. The bracing also featured work holding to suspend an LED engine for concrete embedding.

*Particle-based simulation was used to approximate the rough fluid flows of the dual-gate funnel design, although the result was ultimately a visual exercise rather than comprehensive CFD.



Casting Trials

Failure is a part of the process with artisanal production techniques. A common piece of advice is to not be afraid of failure, but I think that is incorrect. A designer should be afraid of failure, but must fail anyway.



Design For Artisanal Manufacture

Kina was a highly tactile process from design to functional prototyping; however, CAD remained a large part of refining the final form.

Minimising uncertainties, allowing for wide tolerances, and modelling an idealised outcome are all tasks that were needed for the Kina project that CAD does well.

Yield Rate Optimisation

The first three attempts at casting Kina in concrete resulted in de-lamination between the light engine and the concrete form. To fix this, a CNC machinable solution of mechanical keying was developed: an array of partially seated screws.

The dual-gate funnel and LED engine work holding clamp also increased yield rate greatly. These changes improved yield rate from ~10% to ~100% (N=22).

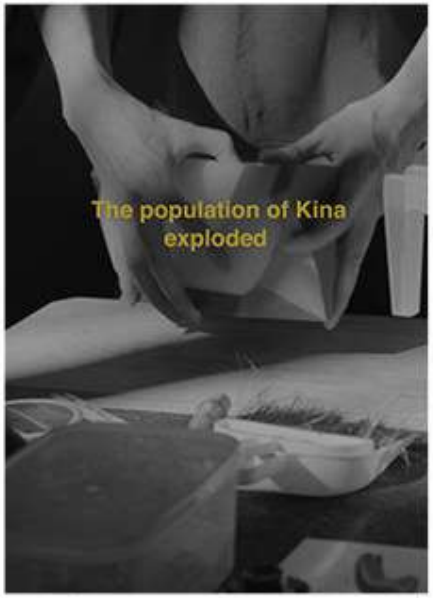






Film About Kina

Documenting the Kina making process.






















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